Computational Neuroscience Study

Neural encoding: Reverse correlation and visual receptive fields

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Outline

Book	Codes
1. Reverse-correlation methods: simple cells	1. Model fitting 2. Machine learning

Reverse-correlation methods: simple cells

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- Spike-triggered average

$$C(x, y, \tau) = \frac{1}{\langle n \rangle} \left\langle \sum_{i=1}^{n} s(x, y, t_i - \tau) \right\rangle$$

Correlation function

$$Q_{rs}(x, y, \tau) = \frac{1}{T} \int_0^T dt\, r(t) s(x, y, t + \tau)$$

- Linear response estimate

$$L(t) = \int_0^\infty d\tau \int dx dy\, D(x, y, \tau) s(x, y, t - \tau)$$

- Space-time receptive field
- Separable receptive field $D(x, y, \tau) = D_s(x, y) D_t(\tau)$
- Nonseparable receptive field

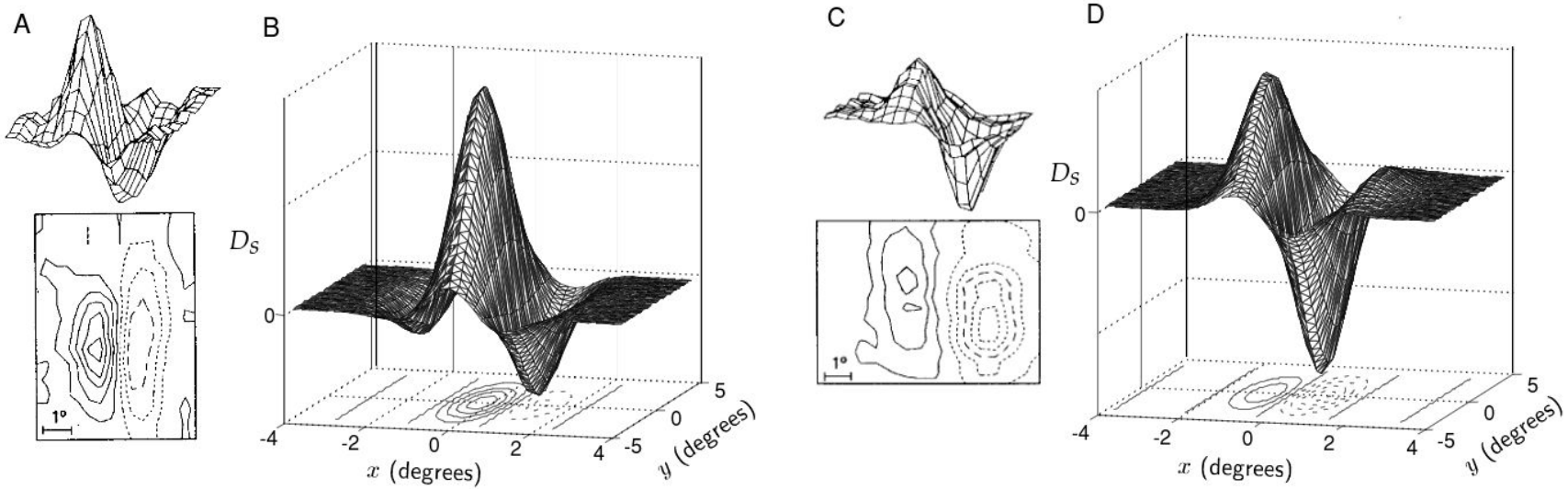
Reverse-correlation methods: simple cells

Spatial receptive fields

- Gabor function (product of Gaussian & sinusoidal function) ⇒ approximates spatial receptive field

$$D_s(x, y) = \frac{1}{2\pi\sigma_x\sigma_y} \exp\left(-\frac{x^2}{2\sigma_x^2} - \frac{y^2}{2\sigma_y^2}\right) \cos(kx - \phi)$$

- k (preferred spatial frequency): spacing of light & dark bars that produces max. response
- ϕ (preferred spatial phase): where ON-OFF boundaries fall within receptive field

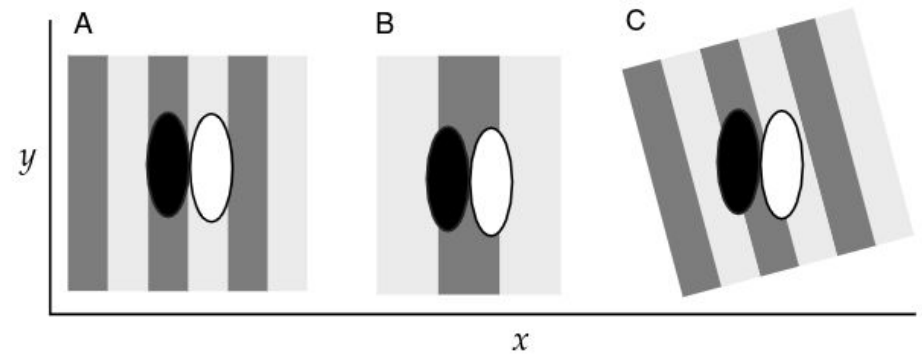


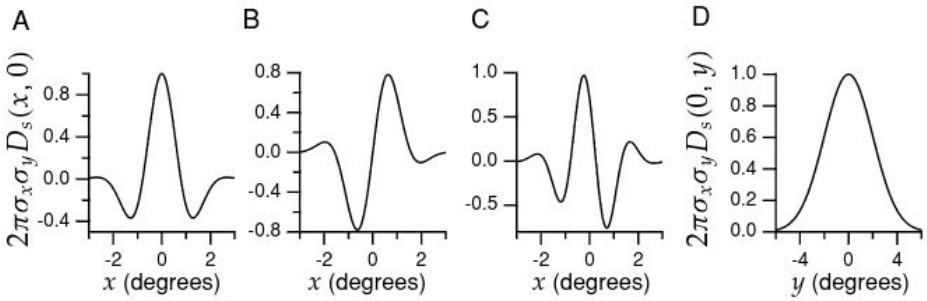
Reverse-correlation methods: simple cells

Spatial receptive fields

- b (bandwidth): number of subregions within a receptive field

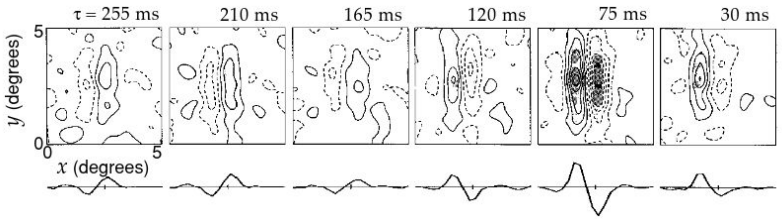
$$b = \log_2\left(\frac{k\sigma_x + \sqrt{2\ln(2)}}{k\sigma_x - \sqrt{2\ln(2)}}\right) \quad \text{or} \quad k\sigma_x = \sqrt{2\ln(2)}\,\frac{2^b+1}{2^b-1}$$





Reverse-correlation methods: simple cells

Temporal receptive fields



- Response of a simple cell to a counterphase grating

$$L(t) = L_s L_t(t) \qquad L_s = \int dx dy\, D_s(x, y) A \cos\left(Kx\cos(\Theta) + Ky\sin(\Theta) - \Phi\right) \qquad L_t(t) = \int_0^\infty d\tau\, D_t(\tau) \cos\left(\omega(t-\tau)\right)$$

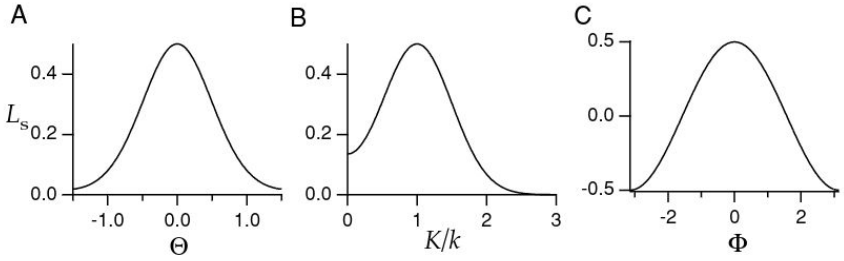


Figure 2.15 Selectivity of a Gabor filter with $\theta = \phi = 0$, $\sigma_x = \sigma_y = \sigma$, and $k\sigma = 2$ acting on a cosine grating with $A = 1$. (A) L_s as a function of stimulus orientation Θ for a grating with the preferred spatial frequency and phase, $K = k$ and $\Phi = 0$. (B) L_s as a function of the ratio of the stimulus spatial frequency to its preferred value, K/k, for a grating oriented in the preferred direction $\Theta = 0$ and with the preferred phase $\Phi = 0$. (C) L_s as a function of stimulus spatial phase Φ for a grating with the preferred spatial frequency and orientation, $K = k$ and $\Theta = 0$.

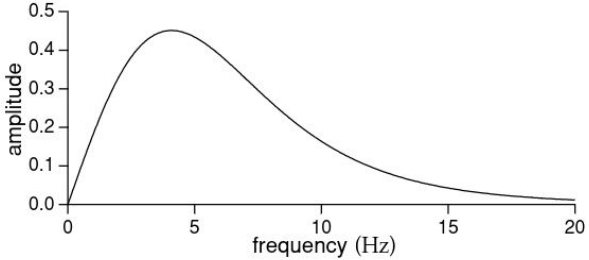


Figure 2.16 Frequency response of a model simple cell based on the temporal kernel of equation 2.29. The amplitude of the sinusoidal oscillations of $L_t(t)$ produced by a counterphase grating is plotted as a function of the temporal oscillation frequency, $\omega/2\pi$.

Reverse-correlation methods: simple cells

Space-time receptive fields

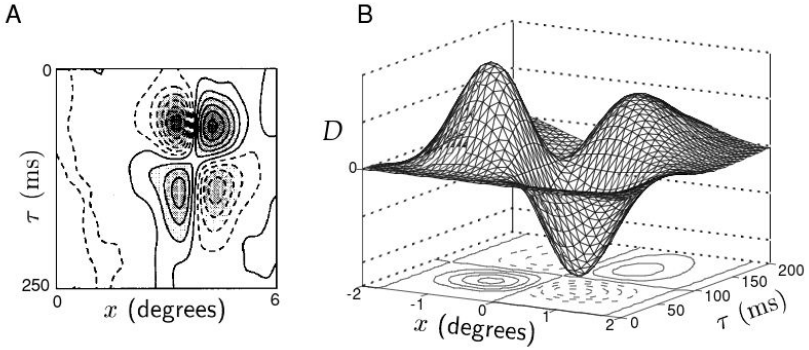


Figure 2.17 A separable space-time receptive field. (A) An x-τ plot of an approximately separable space-time receptive field from cat primary visual cortex. OFF regions are shown with dashed contour lines and ON regions with solid contour lines. The receptive field has side-by-side OFF and ON regions that reverse as a function of τ. (B) Mathematical description of the space-time receptive field in A constructed by multiplying a Gabor function (evaluated at $y = 0$) with $\sigma_x = 1^\circ$, $1/k = 0.56^\circ$, and $\phi = \pi/2$ by the temporal kernel of equation 2.29 with $1/\alpha = 15$ ms. (A adapted from DeAngelis et al., 1995.)

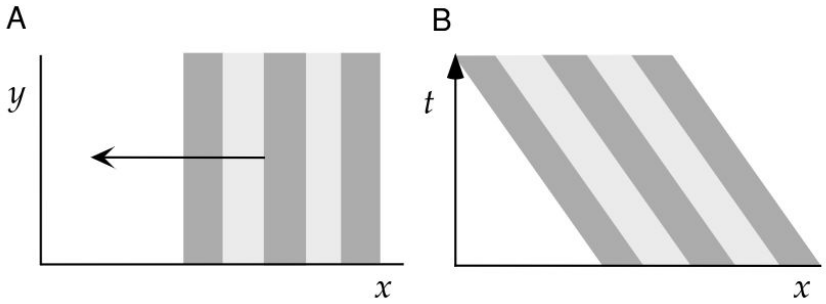


Figure 2.18 Space and space-time diagrams of a moving grating. (A) A vertically oriented grating moves to the left on a two-dimensional screen. (B) The space-time diagram of the image in A. The x location of the dark and light bands moves to the left as time progresses upward, representing the motion of the grating.

Reverse-correlation methods: simple cells

Space-time receptive fields

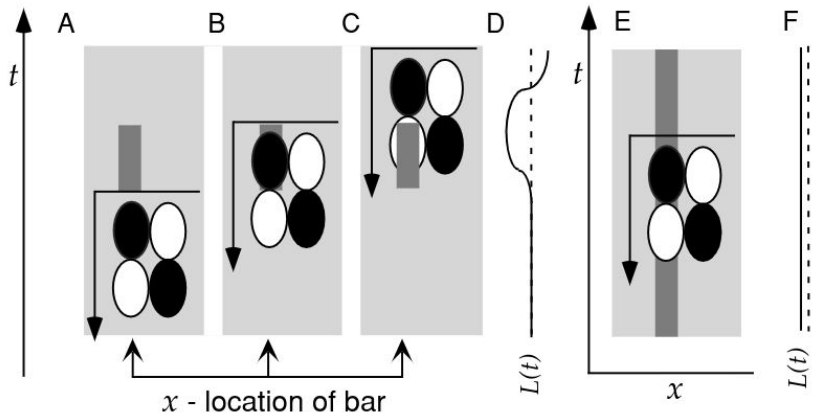


Figure 2.19 Responses to dark bars estimated from a separable space-time receptive field. Dark ovals in the receptive field diagrams are OFF regions and light circles are ON regions. The linear estimate of the response at any time is determined by positioning the receptive field diagram so that its horizontal axis matches the time of response estimation and noting how the OFF and ON regions overlap with the image. (A-C) The image is a dark bar that is flashed on for a short interval of time. There is no response (A) until the dark image overlaps the OFF region (B) when $L(t) > 0$. The response is later suppressed when the dark bar overlaps the ON region (C) and $L(t) < 0$. (D) A plot of $L(t)$ versus time corresponding to the responses generated in A-C. Time runs vertically in this plot, and $L(t)$ is plotted horizontally with the dashed line indicating the zero axis and positive values plotted to the left. (E) The image is a static dark bar. The bar overlaps both an OFF and an ON region, generating opposing positive and negative contributions to $L(t)$. (F) The weak response corresponding to E, plotted as in D.

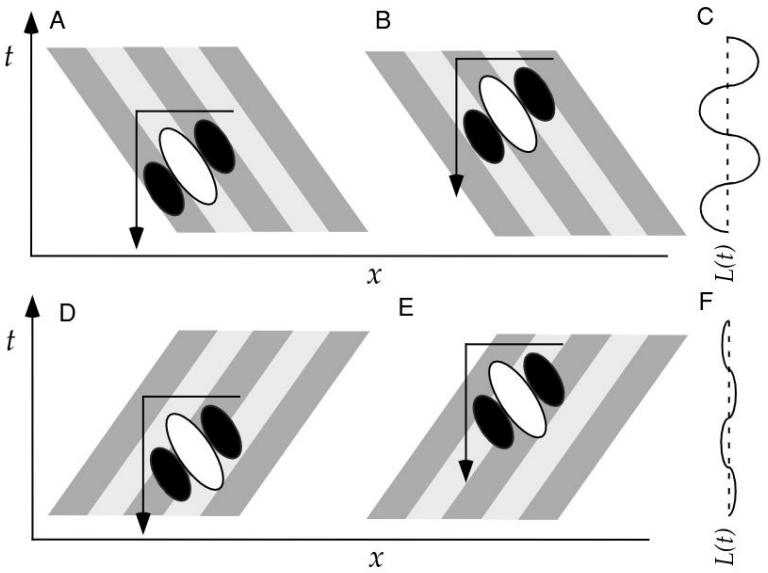


Figure 2.22 Responses to moving gratings estimated from a nonseparable space-time receptive field. Dark areas in the receptive field diagrams represent OFF regions and light areas, ON regions. (A-C) The stimulus is a grating moving to the left. At the time corresponding to A, OFF regions overlap with dark bands and the ON region overlaps a light band, generating a strong response. At the time of the estimate in B, the alignment is reversed, and $L(t)$ is negative. (C) A plot of $L(t)$ versus time corresponding to the responses generated in A and B. Time runs vertically in this plot, and $L(t)$ is plotted horizontally with the dashed line indicating the zero axis. (D-F) The stimulus is a grating moving to the right. Because of the tilt of the space-time receptive field, the alignment with the right-moving grating is never optimal and the response is weak (F).

Reverse-correlation methods: simple cells

Reverse-correlation methods: simple cells

Nonseparable receptive fields

- direction selectivity
- preferred velocity

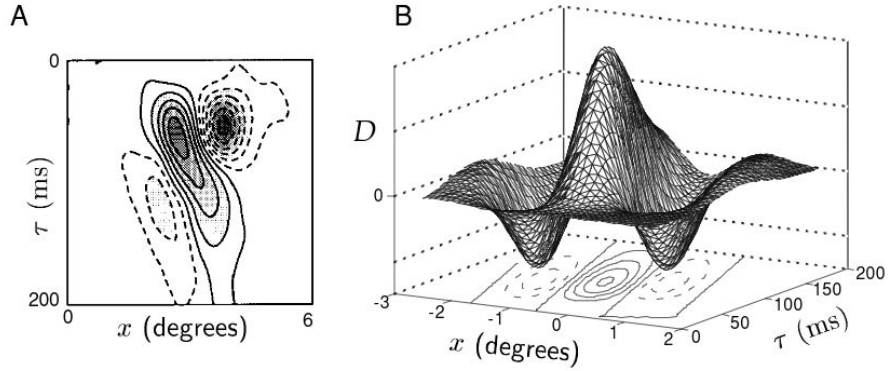


Figure 2.21 A nonseparable space-time receptive field. (A) An x-τ plot of the space-time receptive field of a neuron from cat primary visual cortex. OFF regions are shown with dashed contour lines and ON regions with solid contour lines. The receptive field has a central ON region and two flanking OFF regions that shift to the left over time. (B) Mathematical description of the space-time receptive field in A constructed from equations 2.35 - 2.37. The Gabor function used (evaluated at $y = 0$) had $\sigma_x = 1^\circ$, $1/k = 0.5^\circ$, and $\phi = 0$. D_t is given by the expression in equation 2.29 with $\alpha = 20$ ms, except that the second term, with the seventh power function, was omitted because the receptive field does not reverse sign in this example. The x-τ rotation angle used was $\psi = \pi/9$, and the conversion factor was $c = 0.02\,^\circ/\text{ms}$. (A adapted from DeAngelis et al., 1995.)

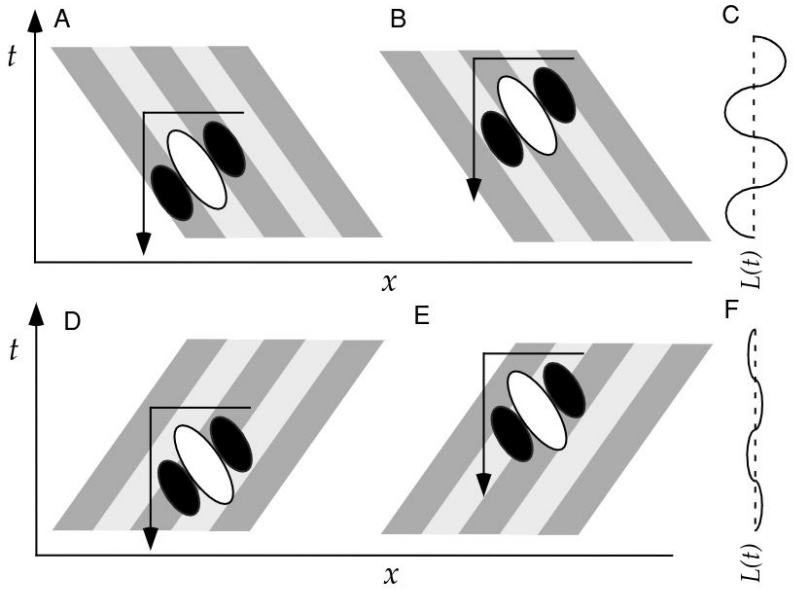


Figure 2.22 Responses to moving gratings estimated from a nonseparable space-time receptive field. Dark areas in the receptive field diagrams represent OFF regions and light areas, ON regions. (A-C) The stimulus is a grating moving to the left. At the time corresponding to A, OFF regions overlap with dark bands and the ON region overlaps a light band, generating a strong response. At the time of the estimate in B, the alignment is reversed, and $L(t)$ is negative. (C) A plot of $L(t)$ versus time corresponding to the responses generated in A and B. Time runs vertically in this plot, and $L(t)$ is plotted horizontally with the dashed line indicating the zero axis. (D-F) The stimulus is a grating moving to the right. Because of the tilt of the space-time receptive field, the alignment with the right-moving grating is never optimal and the response is weak (F).

References

1. Dayan, P., & Abbott, L. F. (2001). Theoretical neuroscience: computational and mathematical modeling of neural systems. Computational Neuroscience Series.